

(ECS708) Report lab6: Neural Networks

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Q1 - Why is it important to use a random set of initial weights rather than initializing all weights as zero in a Neural Network?

It is important to use a random set of initial weights rather than initializing them all to 0, as zero initialization causes all neurons in a layer to behave the same. Every neuron receives the same inputs and produces the same outputs. Therefore, the network doesn't learn diverse features. Whereas random initialization allows each neuron to learn different patterns and for the neural network to learn effectively.

Q2 - How does a NN solve the XOR problem?

Well, as we saw in lab 4 with the XOR problem, logistic regression is a linear classifier that can only learn linear decision boundaries, but the XOR pattern is not linearly separable, as the classes are arranged diagonally. On the other hand, a NN can solve the XOR problem because it introduces nonlinearity and multiple layers.

In the case of XOR, the hidden neurons learn intermediate features, and the output neuron combines these to correctly separate the classes.

Q3 - Explain the performance of the different networks on the training and test sets. How does it compare to the logistic regression example? Make sure that the data you are referring to is clearly presented and appropriately labelled in the report.

Hidden neurons	Final train loss	Final test loss
1	0.4709	0.4652
2	0.2223	0.1905
4	0.1848	0.1581
8	0.1736	0.1482
16	0.1357	0.1129
32	0.1081	0.0908

Table 1: Performance of the different networks

As shown in Table 1, as the number of hidden neurons increases, both the training and test losses decrease. Adding more neurons improves the model's capacity to learn patterns in the data.

In the first network with a single neuron, both the train and test loss are high (0.47), which might suggest underfitting. In other words, the network is too simple to learn the relationships. In contrast, in the networks with 2, 4 and 8 neurons the loss drops significantly, indicating that the model starts to capture more of the information. Finally, the larger networks with 16 and 32 neurons achieve the lowest losses.

Its overall performance is similar to the logistic regression example of lab 4. In that case, the logistic regression was implemented using a one-vs-all strategy, training a separate linear classifier for each class, which allows the model to form multiple linear decision boundaries that approximate a multi-class separation.

The test accuracy in the iris dataset reached 100%, which may be due to the specific train/test split or the fact that the iris dataset is linearly separable, in particular for the Setosa class. However, the neural network by introducing nonlinearity through its hidden layers, is better equipped to model complex relationships and overlapping regions between classes than logistic regression.